Summer Research Project Logbook

Week 1
Day 1 | 24/11/25

- https://www.intel.com/content/www/us/en/support/programmable/support-resources/devices/arria-10-sx-support.html
- Useful resources found for continuing the project such as the FGPA AI Suite and the OpenVINO Toolkit as I will ideally be switching to the Intel Aria 10 board instead of continuing with the cyclone.
- FPGA AI Suite already uses a built in IP for deploying AI inference onto the FPGA but since I already have designed my own custom architecture, I wont exactly be needing that. However, it does provide a more structured end-to-end pathway that may be of use as inspiration, especially on how to use the config files that can generated by the OpenVINO.
- OpenVINO (Open Visual Inference and Neural Network Optimisation) Toolkit is a software suite that is designed to accelerate AI inference across several platforms such as the FPGA that I am using.
    - I would be able to use NIOS II or could even migrate to NIOS V using the Heterogeneous Plugin to delegate my preprocessing to that coprocessor or just make NIOS host the OpenVino Runtime which would essentially become my overall Control Unit, which would manage the memory transactions and instructions for my NPU architecture that I have designed.
    - Would most likely use C++ however, the framework will need to be re-evaluated since AlexNet was used previously on Python script.

Potential Revamps based upon upgrade to Aria 10:

- Number of DSP blocks will be upgraded from 81 to ~1600
- M10K on chip memory to M20K blocks which would an increase from ~6mb to ~48mb

To proceed with the idea of using OpenVINO I would most likely have to use my arch linux OS on my computer to make it work with a C++ based framework instead of the AlexNet that I had on python. For that I would also need to install Intel Quartus Prime Pro

I have found a paper that discusses the analysis of the acceleration of deep learning inference models on a heterogenous architecture based on OpenVINO. This could help justify the use of the OpenVINO software and potentially help in using it.
https://ieeexplore.ieee.org/document/9668607/citations#citations

The paper discusses the use of the intel arria 10 FPGA alongside CPU or CNN inferencing compared to just a standard CPU inference, as that is extremely helpful when operating at the edge. Nothing that I already don’t know. The most important part is the OpenVINO framework. It comprises of two parts: the intermediate representation (IR) which is the generated via Model Optimiser, converting pretrained model into .bin and .xml. .bin is binary file consisting of weights and biases, and .xml shows the structure of the layers and connectivity. The second part is the inference engine that consists of the C++ or python library which would provide the API for reading the intermediate representation.